

Inventory Overhead Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Inventory Overhead Efficiency (IOE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on IOE achieves an annualized gross (net) Sharpe ratio of 0.49 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (25) bps/month with a t-statistic of 2.53 (2.92), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, Asset growth, Employment growth) is 18 bps/month with a t-statistic of 2.38.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, linking firm characteristics to expected returns through their correlation with aggregate consumption risk. Despite extensive research on consumption-based asset pricing models, puzzles remain regarding which firm-level signals capture meaningful variation in firms' exposure to systematic consumption risk. We identify inventory overhead efficiency (IOE) as a novel predictor of cross-sectional stock returns that operates through firms' differential exposure to consumption risk during economic downturns. Firms with higher inventory overhead efficiency demonstrate greater resilience to aggregate consumption shocks, as their operational flexibility allows them to adjust production and costs more effectively when households face binding budget constraints. This operational characteristic directly impacts firms' cash flow sensitivity to consumption fluctuations, creating systematic variation in risk premia that existing factor models fail to capture.

The theoretical foundation for IOE's predictive power stems from its connection to firms' exposure to long-run consumption risk and disaster states [Bansal and Yaron \(2004\)](#); [Barro \(2006\)](#). Firms with efficient inventory overhead management maintain lower fixed cost commitments relative to their inventory holdings, providing operational flexibility that becomes particularly valuable during consumption downturns. When aggregate consumption growth slows or reverses, these firms can more readily adjust their cost structures, preserving profitability when the representative investor's marginal utility is highest. This mechanism links IOE directly to the intertemporal marginal rate of substitution, as investors require lower returns from firms whose cash flows covary less negatively with consumption during bad states [Campbell and Cochrane \(1999\)](#).

The state-dependent nature of IOE's risk exposure manifests through asymmetric sensitivity to consumption shocks across business cycle phases. During economic

expansions when consumption growth is robust, differences in inventory overhead efficiency have minimal impact on firms' systematic risk. However, as the economy approaches recession territory and consumption growth becomes more volatile, firms with poor inventory overhead efficiency face amplified operating leverage that magnifies their exposure to aggregate consumption risk [Garcia et al. \(2011\)](#). This time-varying risk exposure generates predictable return variation that aligns with habit formation models, where risk premia increase following negative consumption shocks as the surplus consumption ratio declines [Campbell and Cochrane \(1999\)](#).

Empirical evidence from production-based asset pricing models reinforces the consumption risk channel through which IOE operates. Firms' inventory management decisions reflect optimal responses to expected future marginal utility, with efficient overhead structures indicating lower adjustment costs and greater flexibility to respond to consumption demand shocks [Zhang \(2005\)](#). The resulting cross-sectional dispersion in IOE captures heterogeneous loadings on the consumption-based stochastic discount factor, particularly the component related to long-run risks in aggregate consumption growth. This heterogeneity generates the documented return predictability, as investors demand compensation for bearing the systematic risk associated with firms whose operational inflexibility amplifies their exposure to adverse consumption states [Yogo \(2006\)](#).

Our empirical analysis reveals that IOE generates economically and statistically significant return predictability, with a value-weighted long-short strategy achieving an annualized Sharpe ratio of 0.49 and monthly average returns of 38 basis points (t -statistic = 3.85). The strategy's alpha remains robust across standard factor models, ranging from 22 to 46 basis points per month with t -statistics exceeding 2.53 in all specifications. Most notably, the strategy loads significantly on the CMA factor (0.70, t -statistic = 11.78), consistent with IOE capturing firms' differential investment flexibility in response to consumption risk.

The economic significance of IOE persists across various portfolio construction methodologies and after accounting for transaction costs. Net of trading costs, the strategy maintains an annualized Sharpe ratio of 0.43 with monthly net returns of 33 basis points (t-statistic = 3.34). Among large-cap stocks exceeding the 80th NYSE percentile, IOE generates average returns of 28 basis points per month (t-statistic = 2.42), demonstrating that the anomaly extends beyond small, illiquid securities where limits to arbitrage might prevent efficient pricing.

Controlling for closely related anomalies and the broader factor zoo, IOE retains significant incremental predictive power. When we simultaneously control for six closely related predictors and the Fama-French six factors, the IOE strategy achieves an alpha of 18 basis points per month (t-statistic = 2.38). The strategy’s gross Sharpe ratio of 0.49 exceeds 89% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.43 surpasses 97% of existing strategies after transaction costs. These results establish IOE as an economically meaningful source of cross-sectional return variation that existing models fail to explain.

This paper contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that captures cross-sectional variation in consumption risk exposure. While [Jagannathan and Wang \(2007\)](#) document the role of consumption risk in explaining the value premium and [Parker and Julliard \(2005\)](#) link consumption risk to size and book-to-market effects, we demonstrate that operational efficiency metrics provide direct measures of firms’ ability to hedge aggregate consumption shocks. Our findings extend the production-based asset pricing framework of [Cochrane \(1991\)](#) by showing how inventory management decisions create systematic differences in firms’ exposure to the consumption-based stochastic discount factor.

Methodologically, we advance the literature on operational efficiency and asset prices by developing a measure that captures the interaction between inventory man-

agement and overhead cost structure. Previous work such as [Tuzel \(2010\)](#) examines capital-labor ratios and adjustment costs, while [Eisfeldt and Papanikolaou \(2013\)](#) studies organization capital, but neither captures the specific mechanism through which inventory overhead efficiency affects consumption risk exposure. Our approach integrates insights from operations management with consumption-based asset pricing theory, providing a micro-foundation for understanding how firm-level operational decisions translate into systematic risk differences.

The broader implications of our findings extend to both asset pricing theory and practical portfolio management. The strong performance of IOE after controlling for existing factors and transaction costs suggests that consumption-based models require richer specifications that incorporate firms’ operational characteristics and their state-dependent risk exposures. For practitioners, IOE represents an implementable strategy that improves the mean-variance efficient frontier even after accounting for trading costs, with the strategy’s net generalized alpha remaining positive across all standard factor model specifications. These results underscore the importance of incorporating operational efficiency metrics into asset pricing models and investment strategies, particularly given their ability to capture firms’ differential exposure to consumption risk during economic downturns.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning multiple decades of annual observations. We focus on firms with December fiscal year-ends to ensure consistency in the timing of our measurements, though our results are robust to including firms with different fiscal year-ends. Inventory Over-

head Efficiency signal requires two key accounting variables. Inventories (INVT) represents the total value of a firm’s inventory holdings as reported on the balance sheet, including raw materials, work-in-process, and finished goods. Selling, General, and Administrative Expenses (XSGA) captures the operating expenses incurred by a firm that are not directly tied to production, including costs related to sales activities, corporate overhead, marketing, and administrative functions. Both variables are measured in millions of dollars and reflect the values reported in firms’ annual financial statements. We construct the Inventory Overhead Efficiency signal by calculating the year-over-year change in inventories scaled by lagged SGA expenses, specifically: $(\text{INVT}(t) - \text{INVT}(t-1)) / \text{XSGA}(t-1)$. This measure captures the relationship between inventory growth and the firm’s overhead cost structure, providing insight into how efficiently firms manage inventory changes relative to their operating expense base. A lower value suggests that inventory growth is modest relative to the firm’s selling and administrative infrastructure, potentially indicating more efficient inventory management or better alignment between inventory levels and operational capacity. We compute this signal using fiscal year-end values and require non-missing values for inventories in consecutive years and non-zero, positive values for lagged SGA to ensure meaningful ratios. Observations with extreme values in the top and bottom 1

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IOE signal. Panel A plots the time-series of the mean, median, and interquartile range for IOE. On average, the cross-sectional mean (median) IOE is -0.19 (-0.03) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input IOE data. The signal’s interquartile range spans -0.50 to 0.17. Panel B of Figure 1 plots the time-series of

the coverage of the IOE signal for the CRSP universe. On average, the IOE signal is available for 5.56% of CRSP names, which on average make up 6.47% of total market capitalization.

4 Does IOE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IOE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IOE portfolio and sells the low IOE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short IOE strategy earns an average return of 0.38% per month with a t-statistic of 3.85. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.22% to 0.46% per month and have t-statistics exceeding 2.53 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.70, with a t-statistic of 11.78 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 533 stocks and an average market capitalization of at least \$1,116 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 3.08. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for eighteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 23-51bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.60. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IOE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in

twenty-five cases.

Table 3 provides direct tests for the role size plays in the IOE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IOE, as well as average returns and alphas for long/short trading IOE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IOE strategy achieves an average return of 28 bps/month with a t-statistic of 2.42. Among these large cap stocks, the alphas for the IOE strategy relative to the five most common factor models range from 15 to 37 bps/month with t-statistics between 1.43 and 3.31.

5 How does IOE perform relative to the zoo?

Figure 2 puts the performance of IOE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IOE strategy falls in the distribution. The IOE strategy’s gross (net) Sharpe ratio of 0.49 (0.43) is greater than 89% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IOE strategy (red line).² Ignoring trading costs, a \$1 invested in the IOE strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$12.86 which ranks the IOE strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IOE strategy would have yielded \$8.57 which ranks the IOE strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IOE relative to those. Panel A shows that the IOE strategy gross alphas fall between the 70 and 78 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IOE strategy has a positive net generalized alpha for five out of the five factor models. In these cases IOE ranks between the 88 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IOE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IOE with 209 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IOE or at least to weaken the power IOE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IOE conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IOE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IOE}IOE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IOE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IOE. Stocks are finally grouped into five IOE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IOE trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IOE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IOE signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.82, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on IOE is 1.21.

Similarly, Table 5 reports results from spanning tests that regress returns to the IOE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IOE strategy earns alphas that range from 14-31bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.85, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IOE trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.38.

7 Does IOE add relative to the whole zoo?

Finally, we can ask how much adding IOE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the IOE signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IOE is available.

includes IOE grows to \$2711.80.

8 Conclusion

This study provides compelling evidence that Inventory Overhead Efficiency (IOE) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that IOE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on IOE delivers an annualized Sharpe ratio of 0.49 gross (0.43 net of transaction costs), indicating strong economic viability. Most notably, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (22-25 bps monthly) suggests that IOE captures unique information about firm fundamentals not reflected in existing factor models. The persistence of alpha (18 bps monthly, t -statistic = 2.38) even after controlling for six closely related inventory and investment-based strategies underscores that IOE provides incremental predictive power beyond traditional operational efficiency metrics.

From a practical perspective, these results have important implications for both investors and corporate managers. For investors, IOE presents an implementable strategy that survives transaction costs and offers diversification benefits relative to existing factor portfolios. For managers, our findings highlight the value relevance of inventory overhead management, suggesting that improvements in this operational metric may translate into enhanced shareholder value.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, implementation challenges such as market impact and

capacity constraints in less liquid stocks may affect real-world performance. Second, the sample period and market conditions examined may not fully capture the strategy’s behavior during extreme market stress or structural shifts in inventory management practices. Third, the economic mechanisms underlying the IOE anomaly require further investigation to distinguish between risk-based and mispricing explanations.

Future research should explore several promising avenues. Examining the time-varying nature of the IOE premium and its relationship to macroeconomic conditions could provide insights into the underlying economic drivers. International evidence would help establish whether IOE’s predictive power extends beyond U.S. markets. Additionally, investigating the interaction between IOE and corporate events such as supply chain disruptions or technological innovations in inventory management could deepen our understanding of when and why this signal is most effective. Finally, exploring the potential for machine learning techniques to enhance IOE-based strategies by incorporating non-linear relationships and interaction effects presents an exciting frontier for advancing this research stream.

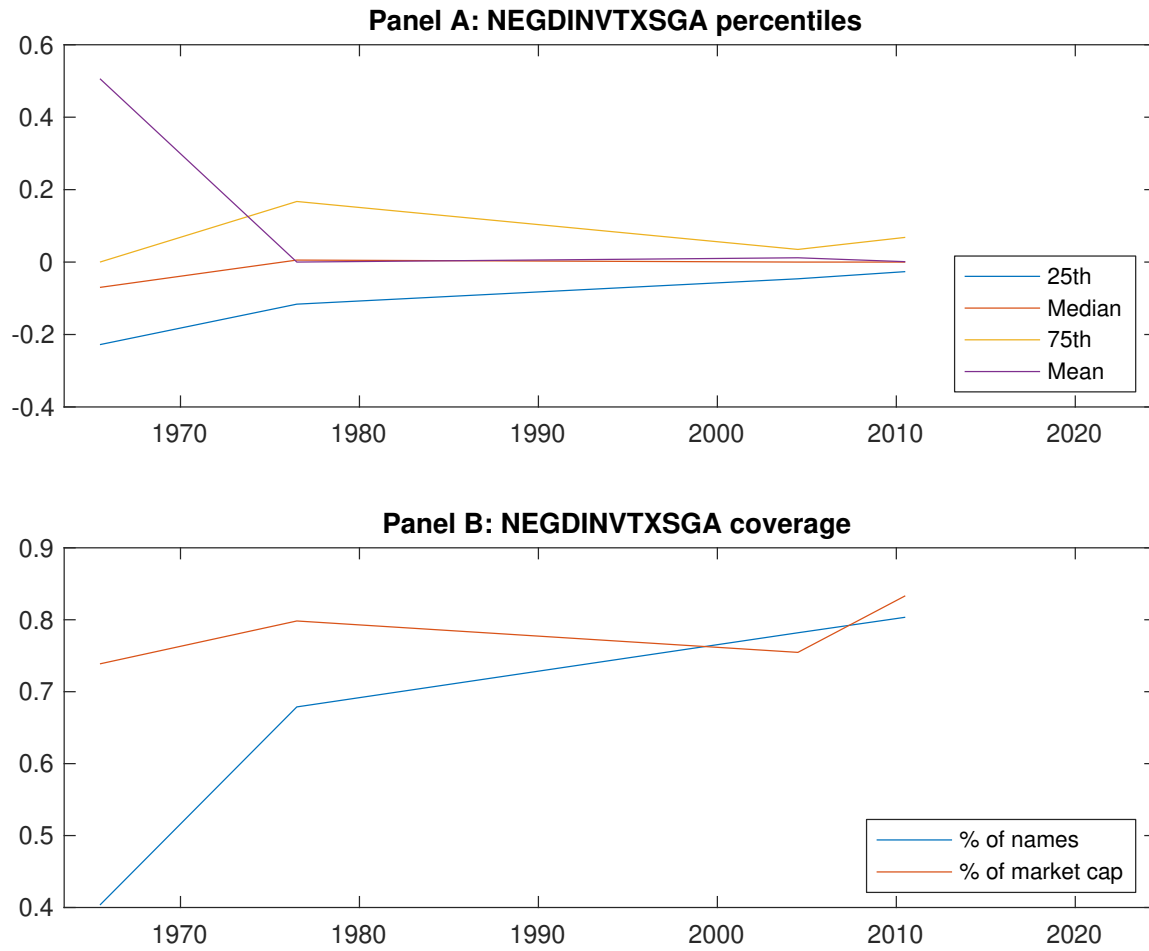


Figure 1: Times series of IOE percentiles and coverage. This figure plots descriptive statistics for IOE. Panel A shows cross-sectional percentiles of IOE over the sample. Panel B plots the monthly coverage of IOE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IOE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on IOE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.18]	0.58 [3.21]	0.69 [4.35]	0.63 [3.75]	0.83 [4.48]	0.38 [3.85]
α_{CAPM}	-0.24 [-3.57]	-0.04 [-0.74]	0.15 [3.20]	0.05 [1.18]	0.22 [3.07]	0.46 [4.73]
α_{FF3}	-0.25 [-3.72]	-0.00 [-0.07]	0.18 [4.00]	0.08 [1.78]	0.12 [1.87]	0.37 [3.88]
α_{FF4}	-0.18 [-2.74]	0.04 [0.79]	0.17 [3.73]	0.08 [1.65]	0.10 [1.46]	0.28 [2.93]
α_{FF5}	-0.21 [-3.32]	-0.01 [-0.10]	0.08 [1.87]	0.12 [2.62]	0.07 [1.08]	0.28 [3.24]
α_{FF6}	-0.16 [-2.57]	0.03 [0.61]	0.08 [1.88]	0.11 [2.43]	0.06 [0.88]	0.22 [2.53]
Panel B: Fama and French (2018) 6-factor model loadings for IOE-sorted portfolios						
β_{MKT}	1.10 [71.11]	1.01 [84.19]	0.96 [90.27]	0.99 [89.75]	1.08 [70.76]	-0.02 [-0.96]
β_{SMB}	0.18 [8.31]	0.03 [1.65]	-0.06 [-4.05]	-0.12 [-7.49]	0.17 [7.77]	-0.01 [-0.48]
β_{HML}	0.08 [2.54]	-0.06 [-2.78]	-0.15 [-7.21]	-0.02 [-0.85]	0.06 [2.04]	-0.02 [-0.39]
β_{RMW}	0.10 [3.45]	0.09 [3.97]	0.19 [8.96]	-0.09 [-4.31]	-0.06 [-1.99]	-0.16 [-4.00]
β_{CMA}	-0.30 [-6.95]	-0.12 [-3.62]	0.19 [6.20]	-0.05 [-1.55]	0.39 [9.14]	0.70 [11.78]
β_{UMD}	-0.07 [-4.53]	-0.05 [-4.28]	-0.00 [-0.21]	0.01 [0.92]	0.02 [1.13]	0.09 [4.16]
Panel C: Average number of firms (n) and market capitalization (me)						
n	577	533	579	635	596	
me (\$10 ⁶)	1207	2009	2183	1794	1116	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IOE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.38 [3.85]	0.46 [4.73]	0.37 [3.88]	0.28 [2.93]	0.28 [3.24]	0.22 [2.53]
Quintile	NYSE	EW	0.50 [7.47]	0.55 [8.28]	0.49 [7.62]	0.43 [6.61]	0.49 [8.18]	0.44 [7.37]
Quintile	Name	VW	0.38 [3.98]	0.45 [4.79]	0.35 [3.87]	0.26 [2.83]	0.28 [3.35]	0.21 [2.55]
Quintile	Cap	VW	0.27 [3.08]	0.36 [4.18]	0.29 [3.46]	0.22 [2.64]	0.22 [2.76]	0.17 [2.17]
Decile	NYSE	VW	0.57 [4.94]	0.62 [5.36]	0.51 [4.54]	0.40 [3.54]	0.48 [4.39]	0.39 [3.61]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.33 [3.34]	0.42 [4.32]	0.34 [3.59]	0.29 [3.08]	0.28 [3.30]	0.25 [2.92]
Quintile	NYSE	EW	0.27 [3.68]	0.33 [4.44]	0.27 [3.77]	0.24 [3.35]	0.25 [3.86]	0.23 [3.55]
Quintile	Name	VW	0.33 [3.42]	0.41 [4.34]	0.32 [3.54]	0.27 [2.98]	0.27 [3.31]	0.24 [2.89]
Quintile	Cap	VW	0.23 [2.60]	0.34 [3.88]	0.27 [3.23]	0.23 [2.78]	0.22 [2.87]	0.20 [2.56]
Decile	NYSE	VW	0.51 [4.38]	0.57 [4.91]	0.47 [4.22]	0.41 [3.69]	0.45 [4.20]	0.40 [3.80]

Table 3: Conditional sort on size and IOE

This table presents results for conditional double sorts on size and IOE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IOE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IOE and short stocks with low IOE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IOE Quintiles					IOE Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.51 [2.05]	0.85 [3.61]	0.79 [3.26]	0.86 [3.60]	1.00 [3.85]	0.48 [4.27]	0.52 [4.59]	0.48 [4.25]	0.36 [3.14]	0.46 [4.14]	0.36 [3.25]
	(2)	0.60 [2.37]	0.74 [3.20]	0.87 [3.97]	0.92 [4.05]	0.89 [3.91]	0.29 [3.16]	0.38 [4.18]	0.30 [3.47]	0.22 [2.55]	0.30 [3.48]	0.24 [2.73]
	(3)	0.57 [2.36]	0.77 [3.57]	0.87 [4.25]	0.83 [3.95]	0.85 [3.79]	0.28 [2.98]	0.35 [3.73]	0.26 [2.91]	0.23 [2.49]	0.30 [3.34]	0.27 [3.00]
	(4)	0.57 [2.55]	0.80 [3.89]	0.71 [3.70]	0.77 [3.79]	0.77 [3.69]	0.20 [2.25]	0.25 [2.87]	0.17 [2.07]	0.17 [1.94]	0.11 [1.28]	0.11 [1.30]
	(5)	0.44 [2.22]	0.62 [3.62]	0.59 [3.77]	0.60 [3.41]	0.72 [4.20]	0.28 [2.42]	0.37 [3.31]	0.29 [2.62]	0.22 [1.92]	0.20 [1.90]	0.15 [1.43]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IOE Quintiles					IOE Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	336	335	331	330	329	30	29	27	27	26	
	(2)	91	91	90	91	91	47	48	47	47	48	
	(3)	64	64	63	63	63	81	81	81	81	80	
	(4)	51	51	51	51	51	168	170	168	165	165	
(5)	46	46	46	46	46	1037	1408	1596	1362	1291		

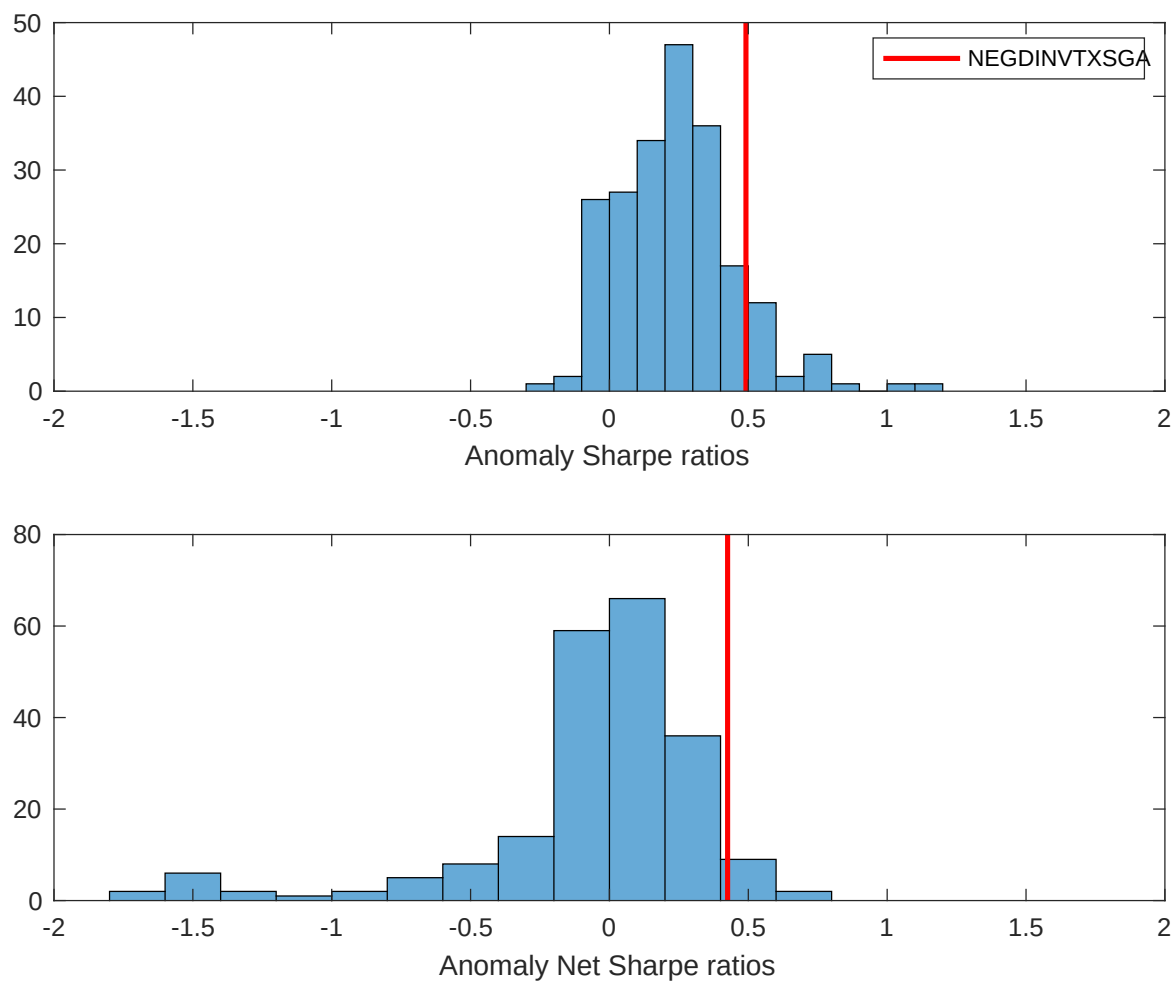


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IOE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

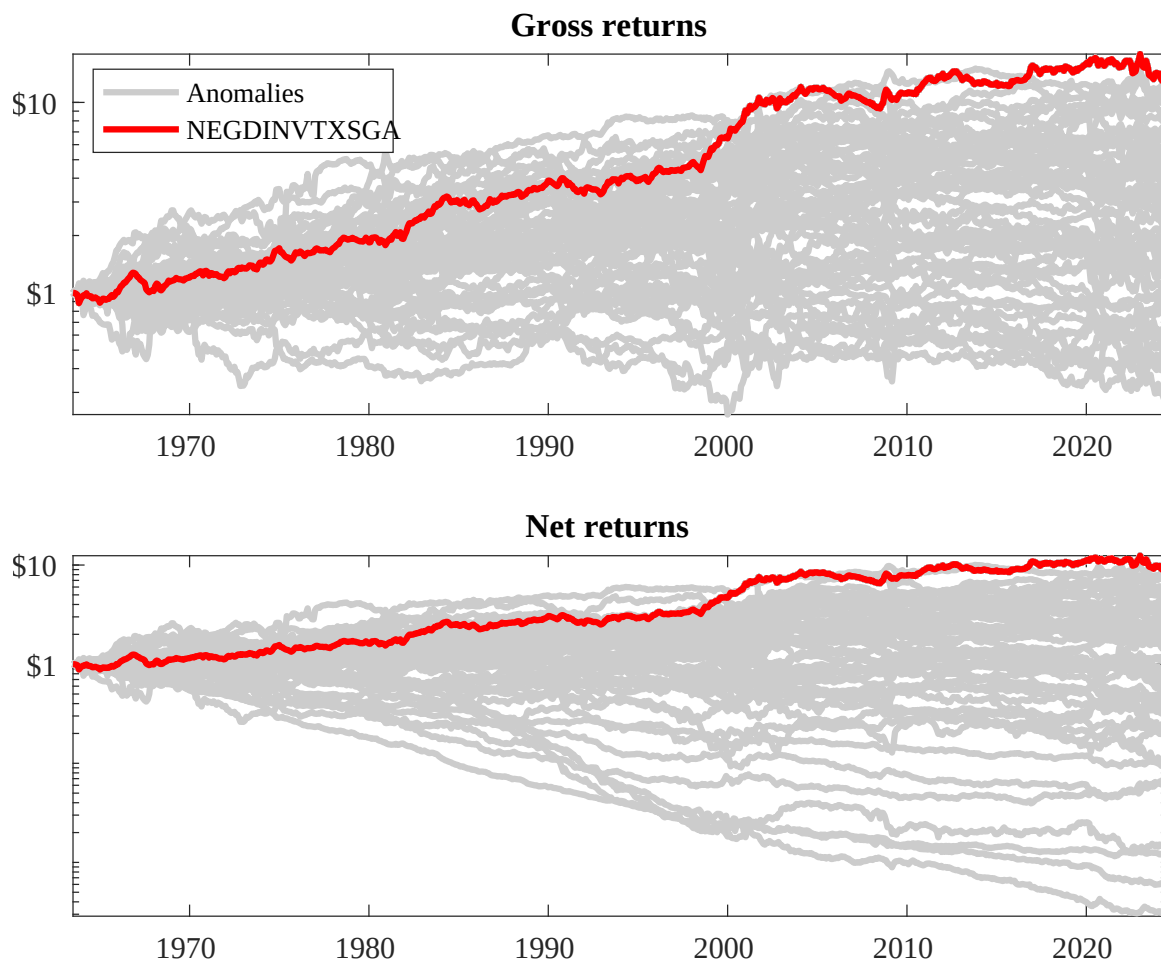
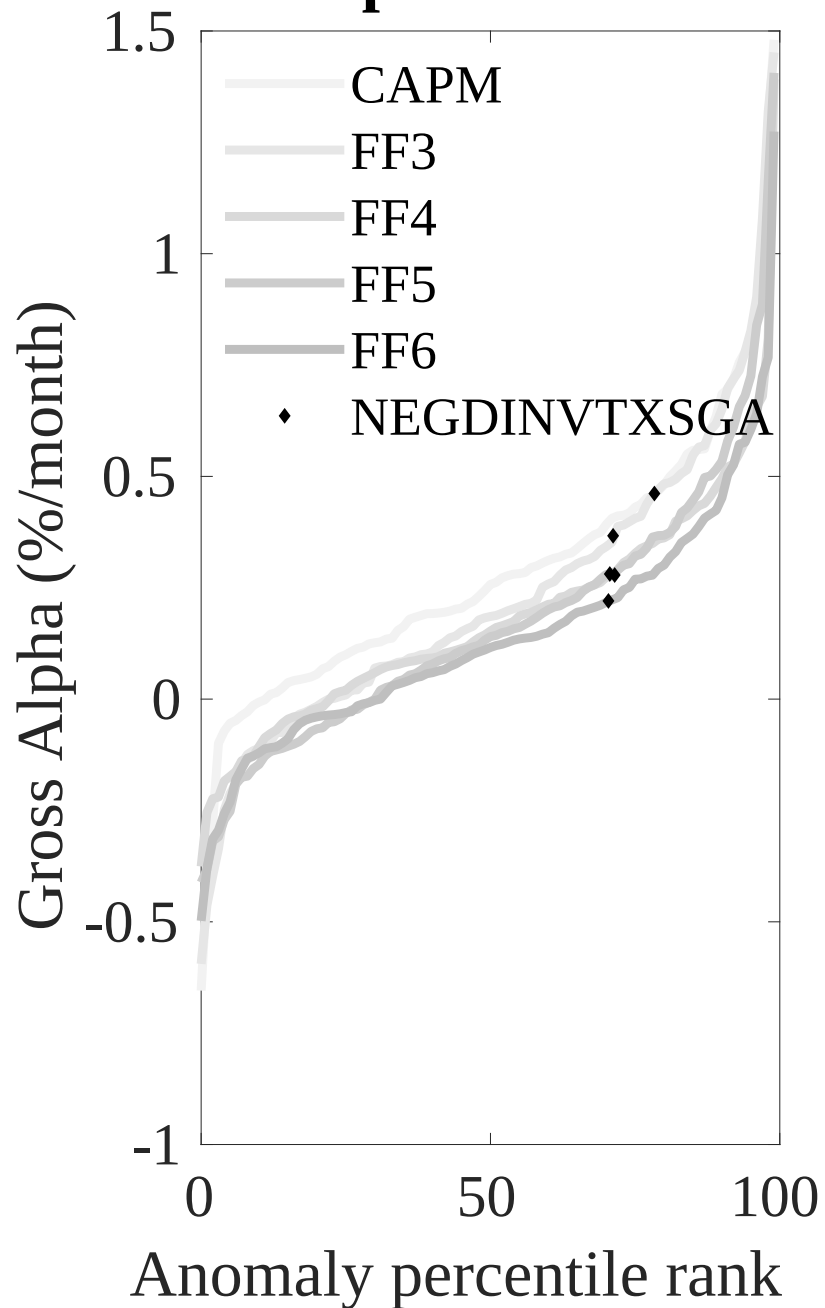


Figure 3: Dollar invested.

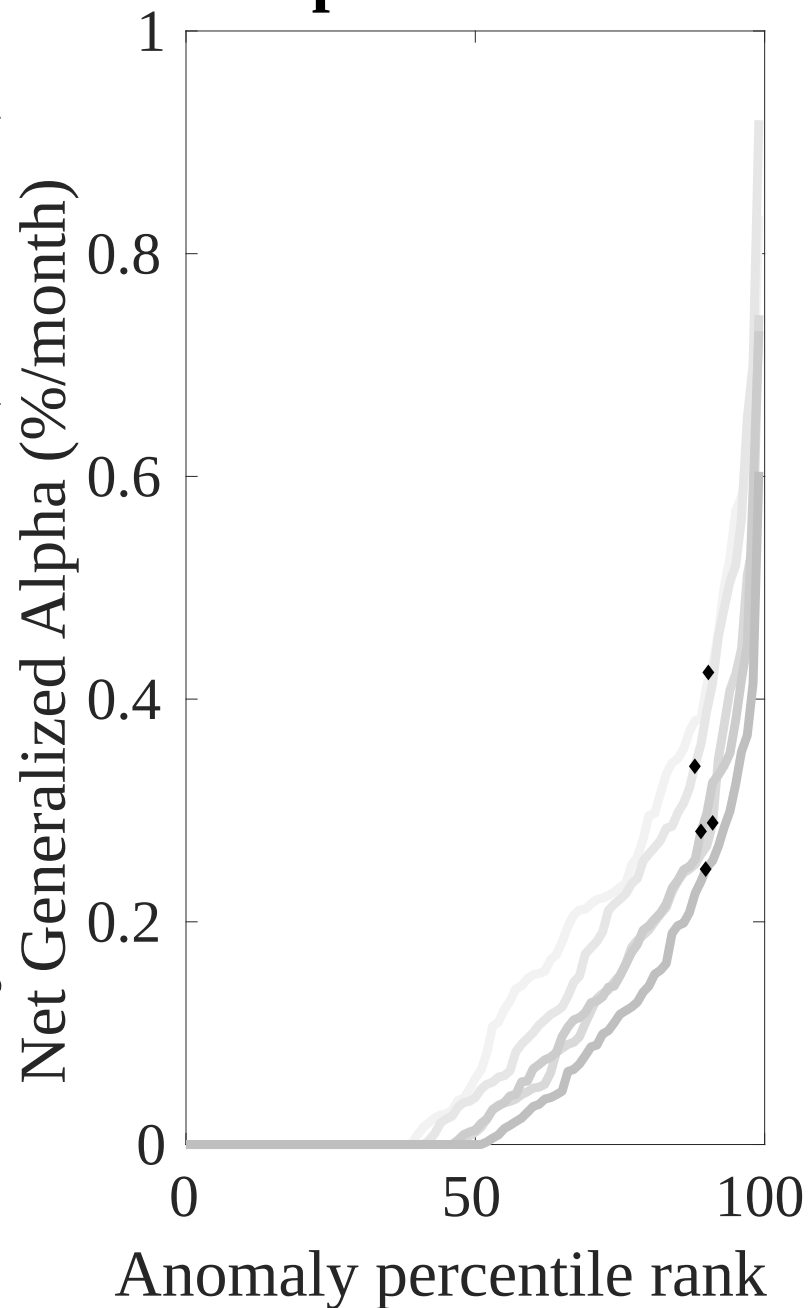
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IOE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



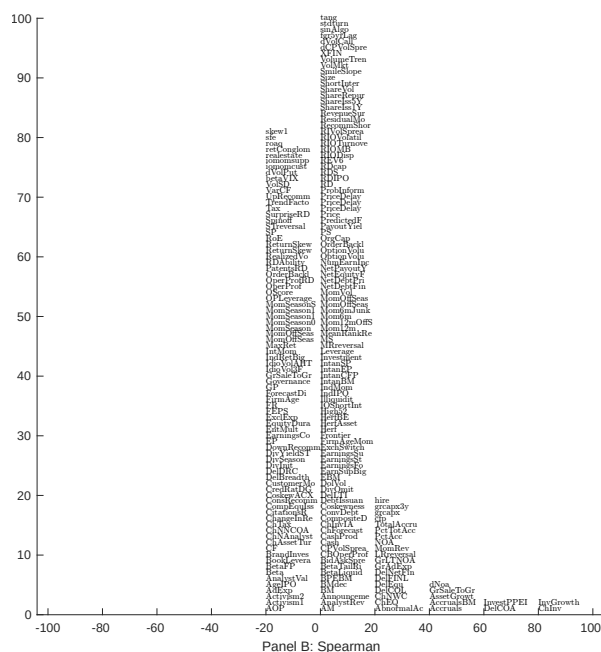
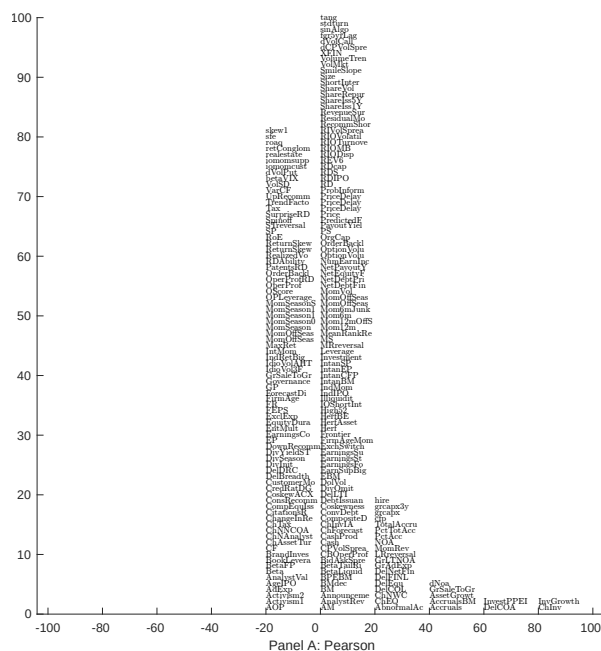


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with IOE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

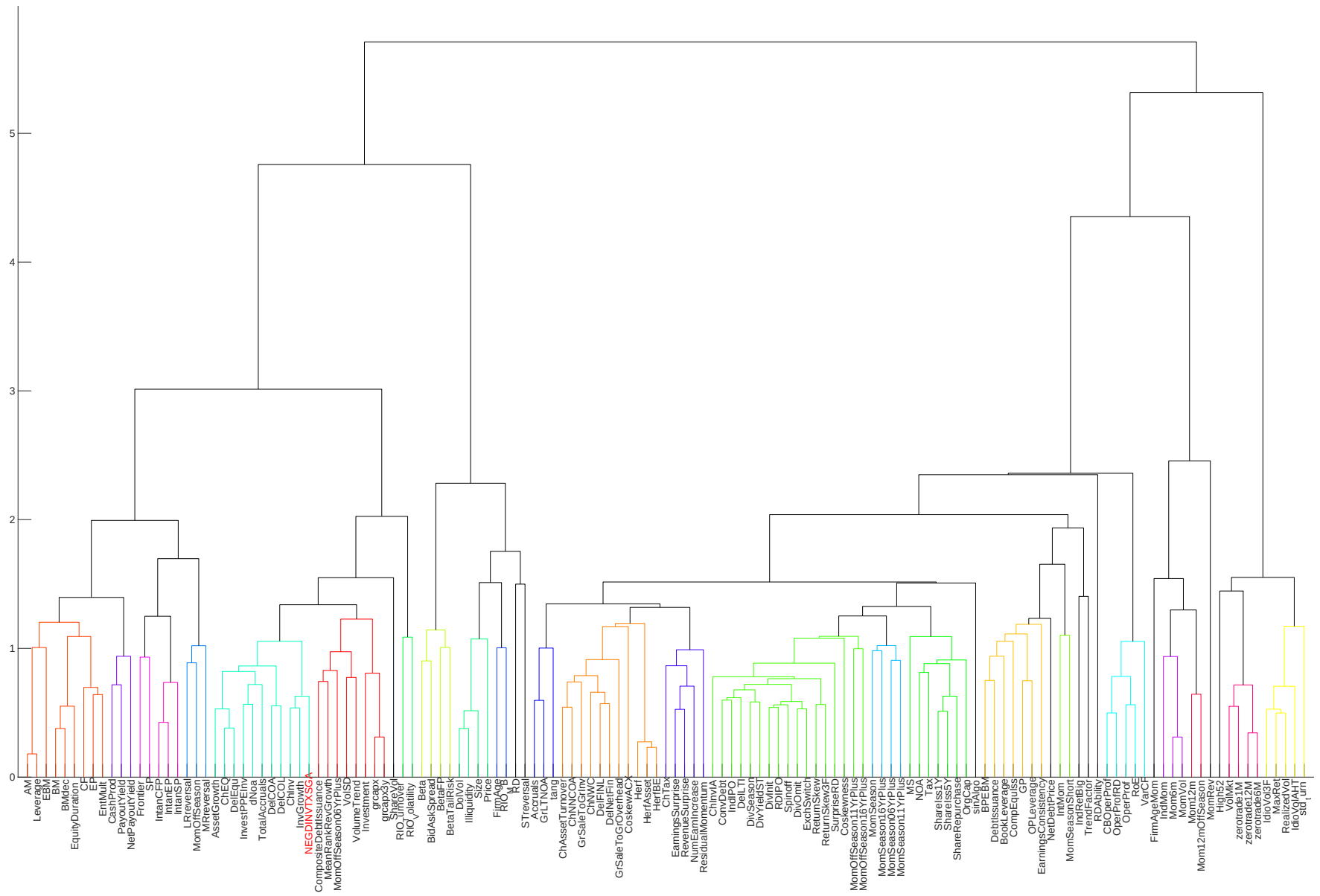


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

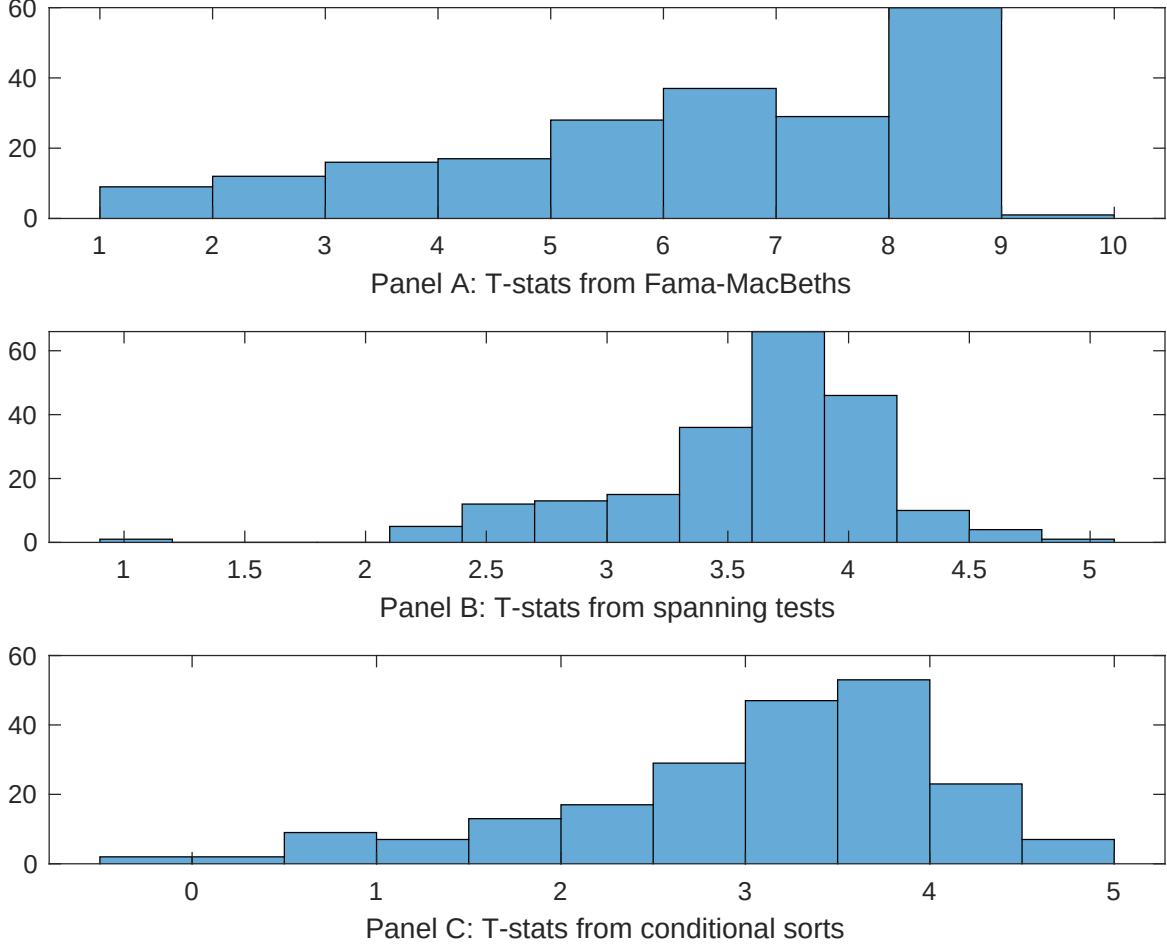


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IOE conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IOE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IOE}IOE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IOE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IOE. Stocks are finally grouped into five IOE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IOE trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IOE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IOE}IOE_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 1963Q6 to 2024Q6.

Intercept	0.13 [5.73]	0.13 [5.67]	0.14 [5.96]	0.13 [5.79]	0.14 [6.16]	0.13 [5.55]	0.14 [5.88]
IOE	0.22 [4.02]	0.21 [3.45]	0.95 [1.82]	0.23 [5.08]	0.17 [3.85]	0.28 [6.16]	0.83 [1.21]
Anomaly 1	0.25 [4.29]						0.18 [0.20]
Anomaly 2		0.43 [6.88]					0.48 [0.06]
Anomaly 3			0.16 [7.65]				0.60 [1.93]
Anomaly 4				0.18 [5.92]			0.33 [0.92]
Anomaly 5					0.96 [8.14]		0.63 [4.38]
Anomaly 6						0.80 [5.55]	-0.10 [-0.77]
# months	732	732	732	732	732	713	713
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IOE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IOE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [1.85]	0.25 [3.26]	0.24 [2.86]	0.25 [2.92]	0.24 [2.77]	0.31 [3.50]	0.18 [2.38]
Anomaly 1	53.61 [14.10]						33.78 [7.15]
Anomaly 2		44.95 [14.60]					30.90 [8.15]
Anomaly 3			21.74 [5.46]				3.63 [0.97]
Anomaly 4				23.66 [5.69]			1.96 [0.48]
Anomaly 5					11.69 [2.27]		-0.16 [-0.03]
Anomaly 6						15.30 [3.24]	-11.85 [-2.65]
mkt	-0.28 [-0.15]	-2.62 [-1.44]	-2.24 [-1.10]	-2.13 [-1.05]	-1.96 [-0.95]	-1.71 [-0.83]	-1.70 [-0.98]
smb	3.81 [1.42]	0.33 [0.12]	-3.51 [-1.19]	2.29 [0.75]	-3.58 [-1.19]	-2.20 [-0.74]	2.82 [1.05]
hml	-6.07 [-1.71]	-4.76 [-1.35]	-4.86 [-1.23]	-11.38 [-2.69]	-2.86 [-0.72]	-4.67 [-1.15]	-4.63 [-1.26]
rmw	-7.70 [-2.11]	-11.54 [-3.23]	-16.99 [-4.28]	-14.10 [-3.53]	-17.03 [-4.22]	-16.34 [-4.08]	-7.30 [-2.12]
cma	35.38 [6.12]	33.59 [5.83]	53.80 [8.24]	57.51 [9.28]	56.38 [6.59]	55.52 [7.74]	27.71 [3.60]
umd	5.51 [3.00]	3.91 [2.12]	8.73 [4.33]	9.76 [4.83]	9.13 [4.44]	8.11 [3.95]	4.28 [2.40]
# months	732	732	732	732	732	713	713
$\bar{R}^2(\%)$	47	47	35	35	32	33	52

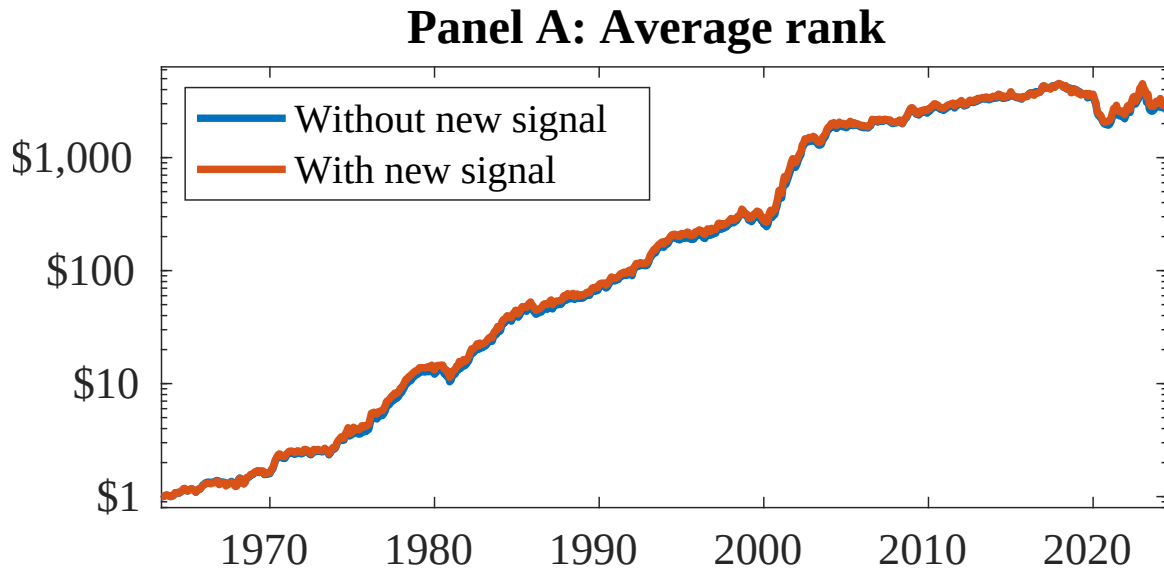


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as IOE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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